KLAi

Regression: explaining and predicting a number

How much does one thing change when another changes, and how well can a model predict it?

AI and Data Science Professional | Session 13 of 24 | Module C: Analyse and explain
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Fit a simple and a multiple linear regression by ordinary least squares, and read each coefficient as a sentence a manager can use



Judge a model with residuals, R squared and a residual plot, and put a category into a model with indicator variables



Say what a regression can predict, what it can explain, and what it cannot prove

Your two hours: concepts first, then a Python regression lab

0:00	10	Recall and the question of the day
0:10	45	Concepts: picture, plain words, one example by hand
0:55	10	Break
1:05	45	Lab: predict, run, read, change
1:50	10	Exit check, keep and next

Ideas with pictures first. Then you open the notebook, fit the model and write your one page report.

Three questions from last time

1. A 95 percent confidence interval for a lift includes zero. What do you tell the manager?

2. A p-value is tiny. Does that make the effect big?

3. What does the standard error measure?



Answer aloud first. Then we check together.

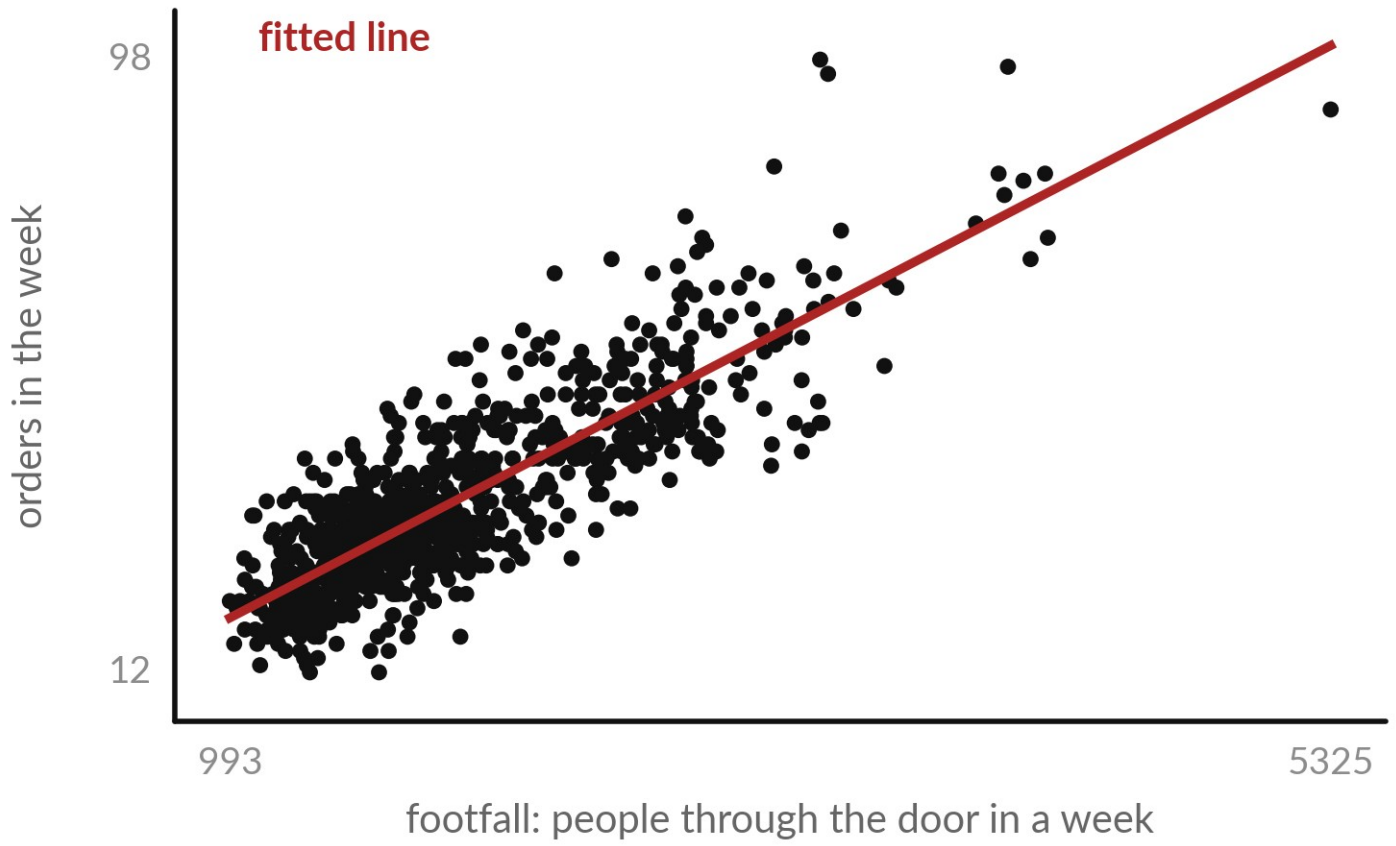
Answers to the three questions from Session 12

	Question	Short answer
	**A 95 percent confidence interval for a lift includes zero. What do you tell the manager?**	We cannot tell yet whether the lift is more than sampling variability. The data are compatible with a gain, with no change and with a loss.
	**A p-value is tiny. Does that make the effect big?**	No. The p-value says a gap this size is unlikely if nothing is going on. The effect size says whether the gap matters.
	**What does the standard error measure?**	How much a statistic, such as a sample mean, varies from one sample to the next. It shrinks as the sample grows.

Words we use today

	Word	What it means
	**Regression (linear regression model)**	A model that predicts a numeric target as an intercept plus a weighted sum of predictors. Ordinary least squares (OLS) finds the weights.
	**Coefficient and intercept**	The coefficient is the change in the prediction for one more unit of a predictor, with the other predictors held fixed. The intercept is the prediction when every predictor is 0.
	**Residual (prediction error)**	Actual value minus predicted value for one record. OLS minimises the sum of the squared residuals.
	**R squared (coefficient of determination)**	The share of the variation in the target that the model accounts for, from 0 to 1.
	**Indicator variable (dummy variable)**	A variable that is 1 when a record belongs to a category and 0 otherwise. One-hot encoding makes one for each level.

Regression fits a line through a cloud: weekly orders against footfall



A scatter plot has one dot for each record: one branch in one week.

832 dots, one for each record.

Along the bottom: footfall, the predictor (independent variable).

Up the side: orders, the target (dependent variable).

The Excel word, the statistics word and the machine learning word for each part

	In Excel	In statistics	In machine learning
	**Trendline on a scatter chart**	Linear regression model	Model (regressor)
	**The x column**	Independent variable (predictor)	Feature
	**The y column**	Dependent variable (response)	Target
	**SLOPE and INTERCEPT**	Coefficient and intercept	Parameters (weights and bias)
	**RSQ, or R2 on the trendline**	R squared	Score (model.score)
	**A row**	Observation	Instance (record)

LINEST is the Excel function that fits several predictors at once. This course uses predictor and target from here on.

Simple linear regression is a formula with two numbers: an intercept and a coefficient

Simple linear regression

$$\hat{y} = b_0 + b_1 x$$

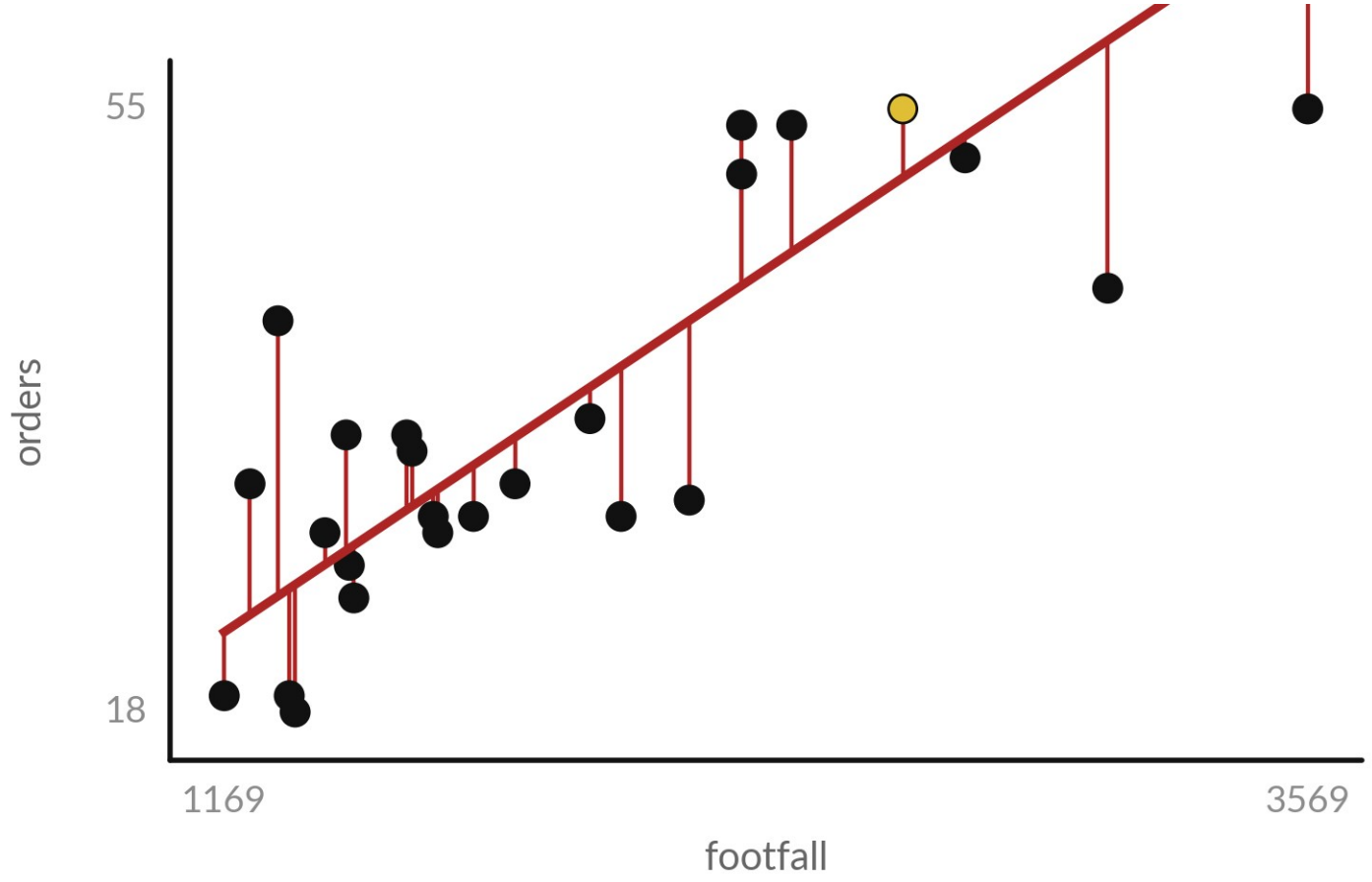
Our fitted model

$$\hat{y} = 1.23 + 0.0185 x$$

Symbol	What it means	In our data
y-hat	The predicted target.	orders in a week
x	The predictor.	footfall, 993 to 5,325 people
b0 (intercept)	The prediction when x is 0.	1.23 orders
b1 (coefficient)	The change in the prediction for one more unit of x.	0.0185 orders per person

1,000 more people at the door goes with about 18.5 more orders. R squared is 0.706, the correlation 0.84 squared.

A residual is the actual value minus the predicted value



Each red segment is one residual: how far the line misses one record.

Residual: e = y minus y-hat. Above the line it is positive.

Gold dot: B01, week of 2024-01-01. Actual 55 orders, predicted 50.8, residual +4.2.

25 of the 832 records are drawn.

Ordinary least squares picks the line with the smallest sum of squared residuals

Residual of record i

$$e_i = y_i - \hat{y}_i$$

Ordinary least squares

$$\min_{b_0,\ b_1} \sum_{i=1}^{n} e_i^2$$

Symbol	What it means	In our data
e_i	The residual of record i.	+4.2 for record 1
y_i and y-hat_i	Actual and predicted target of record i.	55 and 50.8
n	The number of records.	832
sum of e_i squared	Add the squared residuals of all n records.	43,930 for our line

A rival line with slope 0.015 (and its own best intercept) gives a sum of 47,780. OLS finds the smallest sum: 43,930.

Multiple linear regression adds predictors: footfall, promotion and rainfall

Multiple linear regression

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3$$

Our fitted model

$$\hat{y} = 2.38 + 0.0188\, x_1 + 6.80\, x_2 - 0.118\, x_3$$

Symbol	What it means	In our data
x1	footfall, people through the door.	0.0188 orders per person
x2	promo_on: 1 in a promotion week, 0 otherwise.	6.80 orders
x3	rainfall_mm, millimetres of rain in the week.	-0.118 orders per mm
b0	The intercept.	2.38 orders

One least squares rule finds all four numbers. statsmodels and scikit-learn give the same numbers as NumPy.

Read each coefficient as one sentence, with its 95 percent confidence interval

	Term	Coefficient	95% interval	As a sentence
	**Intercept**	2.38	0.91 to 3.84	The starting level. No week has zero footfall (the minimum is 993), so do not read it alone.
	**footfall**	0.0188	0.0181 to 0.0195	1,000 more people at the door goes with 18.8 more orders, other predictors held fixed.
	**promo_on**	6.80	5.79 to 7.80	A promotion week has 6.8 more orders than a week without one at the same footfall and rain.
	**rainfall_mm**	-0.118	-0.136 to -0.100	10 mm more rain goes with 1.2 fewer orders, other predictors held fixed.

Interval = coefficient plus or minus 1.963 standard errors (Student t, 828 degrees of freedom).

R squared is the share of the variation in the target that the model accounts for

R squared

$$R^2 = 1 - \frac{\sum e_i^2}{\sum (y_i - \bar{y})^2}$$

Our model

$$R^2 = 1 - \frac{30{,}660}{149{,}301} = 0.795$$

Symbol	What it means	In our data
sum of e_i squared	Sum of squared residuals: what the model leaves unexplained.	30,660
sum of (y_i - y-bar) squared	Total variation of the target around its mean y-bar.	149,301
Adjusted R squared	R squared with a penalty for each extra predictor.	0.794

Footfall alone gave 0.706. RMSE (6.1 orders) is the typical miss, weighting big misses more.

A category enters a regression as indicator (dummy) variables: one-hot encoding

Before: one categorical variable

week_start	branch_id
2024-01-15	B01
2024-01-15	B03
2024-01-15	B02
2024-01-15	B06



After: one indicator (dummy) column per level

B01	B02	B03	B04	B05	B06	B07	B08
1	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0
0	0	0	0	0	1	0	0

B01: dropped

One-hot encoding keeps 7 of the 8 columns. The dropped level (B01) is the reference level. A record with all 7 kept indicators at 0 belongs to B01.

promo_on is already an indicator variable. branch_id has 8 levels, so we make 8 columns and keep 7.

Keep all eight indicators and the intercept and the model cannot be solved: the dummy variable trap

Design matrix for three records (one row each)

intercept	B01	B02	B03	B04	B05	B06	B07	B08	sum
1	1	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	0	0	0	1

The intercept column is always 1. The indicators add up to 1 in every row, so they repeat it.

9 columns, but the matrix rank is only 8: there is no single best set of coefficients.

This is perfect multicollinearity. The fix is one-hot encoding with one level dropped: 8 columns, rank 8.

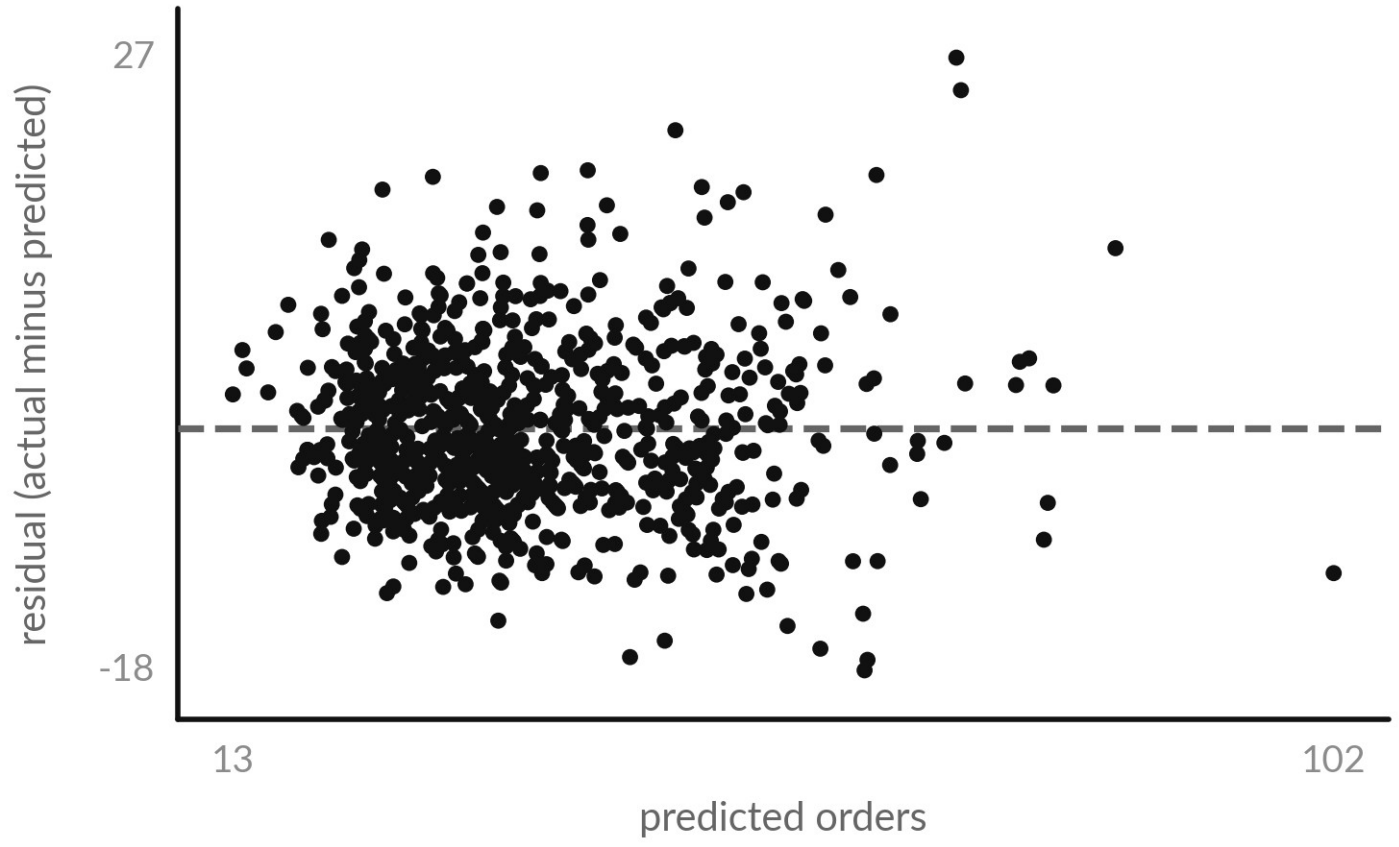
Perfect multicollinearity: one column is a copy of the sum of others.

Seven branch indicators barely change the fit: R squared 0.796, and adjusted R squared falls

	Model	Predictors	R squared	Adjusted R squared
	**Footfall only**	1	0.706	0.705
	**Footfall, promotion, rainfall**	3	0.795	0.794
	**Plus 7 branch indicators (B01 is the reference level)**	10	0.796	0.793

R squared never falls when you add a predictor. Adjusted R squared falls when the new predictor adds too little. At the same footfall, promotion and rain, B02 has 0.82 fewer orders than B01.

A residual plot should show an even cloud: homoscedasticity



Homoscedasticity: the residuals have the same spread at every prediction.

Ours grows: standard deviation 5.2, 5.6, then 7.2 orders, low to high thirds.

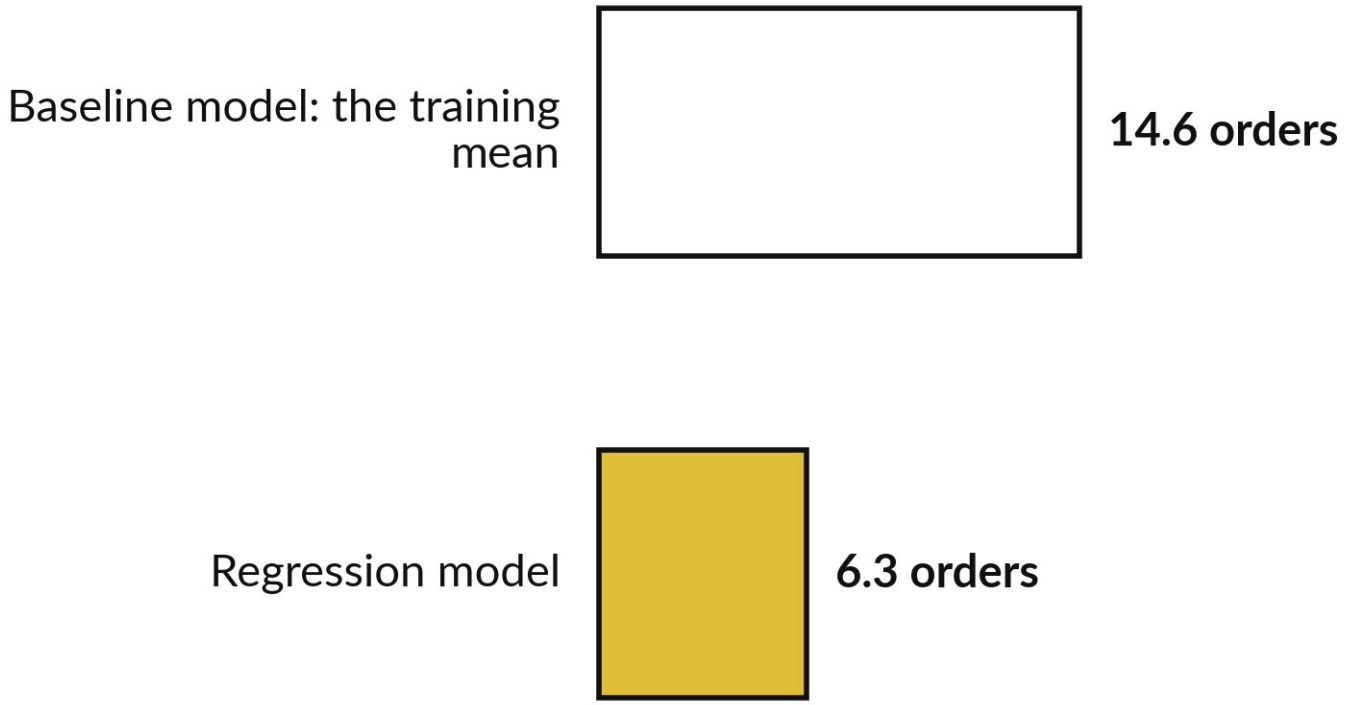
A fan shape: mild heteroscedasticity.

Intervals for busy weeks are too narrow.

Four patterns in a residual plot and what to do about each

	Pattern	What it means	What to try
	**An even cloud around zero**	The straight-line model and constant spread look reasonable.	Report the model with its limits.
	**A curve, like a smile**	The relationship is not a straight line.	Add a squared term or change the variable.
	**A fan that widens (heteroscedasticity)**	The error grows with the size of the prediction.	Model the log of the target, or report wider intervals for large predictions.
	**A few dots far from the rest**	Outliers or data errors.	Check those records. Do not delete without a reason.

Prediction: on 208 records it never saw, the regression misses by 6.3 orders and the baseline model by 14.6



RMSE on the test set: the typical miss on weeks the model never saw.

Training set: 624 records, 2024-01-01 to 2025-06-23.

Test set: 208 records, 2025-06-30 to 2025-12-22.

Baseline model: predict the training mean (35.7) every time.

Explanation and prediction are two jobs with two ways to check the model

Explanation

How much does the target change when a predictor changes?

Read the coefficients and intervals.

Check the residual plot.

Prediction

What will the target be for a new record?

Score the model on records it never saw.

Beat the baseline model.

Say which job you are doing before you fit the model.

A coefficient depends on the other predictors in the model: promotion is 5.61 alone and 6.80 with footfall and rain

	Model	promo_on coefficient	rainfall_mm coefficient
	**promo_on alone**	5.61	not in the model
	**promo_on, footfall and rainfall_mm**	6.80	-0.118
	**rainfall_mm alone**	not in the model	-0.125

Promotion weeks had slightly lower footfall (1,833 people against 1,910), so footfall hides part of the promotion effect. Rain and footfall are almost unrelated (correlation -0.004), so the rain coefficient hardly moves.

A coefficient shows an association, and a cause needs more evidence

What the model supports

Promotion weeks go with 6.8 more orders at the same footfall and rain.

Predicting a planned week.

What it cannot show

That the promotion caused the extra orders.

Nobody randomised the promotion weeks, so a confounder could explain part of the gap.

A cause needs an experiment or strong evidence.

Predict one week by hand: 2,000 people, a promotion and 20 mm of rain gives 44.3 orders

Worked example with the fitted multiple linear regression model.

Question	A branch expects 2,000 people, a promotion week and 20 mm of rain. How many orders?
Working	`2.378 + 0.01877 x 2,000 + 6.799 x 1 - 0.1183 x 20 = 44.35`
Answer	About 44 orders. The same week without a promotion gives 37.5.
Check	2,000 lies inside the footfall range of the data (993 to 5,325). Outside that range the prediction is an extrapolation.

Where you will use this at work: put a number on what moves your target

	Your job	The question you can answer	The tool you would use
	**Retail analyst at a supermarket chain in Kampala**	How many more orders does a promotion week bring at the same footfall?	Excel LINEST, or statsmodels in Python
	**Planning officer at a district health office**	How do monthly clinic visits change with population and the season?	pandas and scikit-learn, results shared in Excel
	**Operations manager at a coffee cooperative**	Illustrative: how do kilograms delivered change with rainfall and price?	Excel trendline first, then Python
	**Finance analyst at a bank**	What explains branch deposits? One page of coefficients for the credit committee.	statsmodels summary, Power BI for the chart

A good project question names the target, the predictors, the data and the decision

	Weak question	Better question	What it names
	**Is our data interesting?**	Which of footfall, promotion and rainfall best explain weekly orders at Lakeside Mart branches?	Target and predictors
	**Can we predict loans?**	Can we predict which Nile Microfinance loans default, using only what is known when the loan is approved?	Target, data, timing
	**Will this help clinics?**	How many visits should a district clinic plan for next month, and how wrong is the forecast likely to be?	Decision and error
	**My own question**	What will I predict or explain, from which table, to help which decision?	Your turn

Read the recipe before the lab: four lines fit a multiple regression

```
cols = ['footfall', 'promo_on', 'rainfall_mm']
X = np.column_stack([np.ones(len(bw)), bw[cols]])
coef, *_ = np.linalg.lstsq(X, bw.orders, rcond=None)
residuals = bw.orders - X @ coef
print(*coef.round(4))
```

```
2.3785 0.0188 6.7986 -0.1183
```

In plain words

Line 1 names the three predictors.

Line 2 adds a column of ones, which carries the intercept, to make the matrix X.

Line 3 solves ordinary least squares. coef holds the intercept, then one coefficient for each predictor.

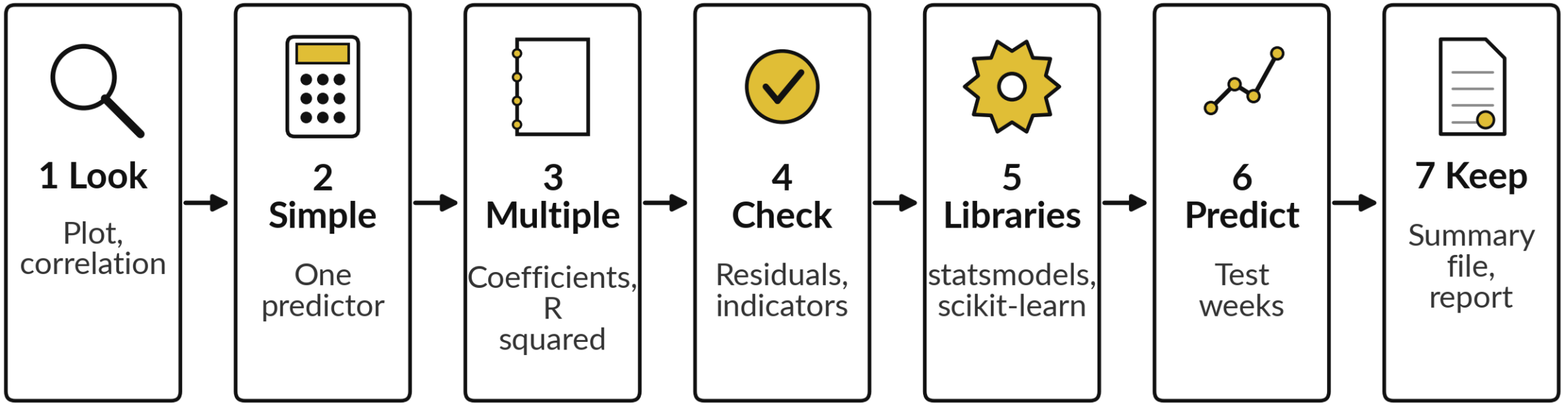
Line 4: the residuals are actual minus predicted.



Part 2: the regression lab

Close the slides. Open the notebook.

The lab has seven steps and ends with a one page report and a summary file



Predict, run, read, change in each step. Checks tell you when your numbers match.

Check your numbers before you move on

What you compute	You should get
Records, and the correlation of orders with footfall	832, and 0.84
Simple regression: intercept and coefficient of footfall	1.23 and 0.0185
Multiple regression: intercept, footfall, promo_on, rainfall_mm	2.38, 0.0188, 6.80, -0.118
R squared, and the RMSE (a typical size of a miss)	0.795 and 6.1 orders
Residual standard deviation, lowest and highest third of predictions	5.2 and 7.2
R squared and adjusted R squared with 7 branch indicators	0.796 and 0.793
Matrix rank with the intercept and all 8 indicators	8 (9 columns)
Test RMSE: baseline model and regression	14.6 and 6.3

If your number differs, find the step that went wrong. Do not change a number to match.

Five likely mistakes in the lab and how to fix them

	What you see	What it means	What to do
✗	**The intercept is missing, or the numbers are far from the table**	X has no column of ones.	Add np.ones(len(bw)) as the first column of X.
✗	**A ValueError about shapes**	X and y have different lengths, or a column is missing.	Print X.shape and y.shape. The rows must match.
✗	**Huge coefficients, or a statsmodels multicollinearity note**	All 8 indicators and the intercept: the dummy variable trap.	Use drop_first=True. Check np.linalg.matrix_rank.
✓	**The residual plot shows a wedge**	That is the heteroscedasticity from the residual plot slide.	It is a finding. Write it in your report.
✗	**statsmodels gives a column named Intercept**	The formula adds an intercept for you.	Do not add a column of ones in the formula version.

Your Project milestone 2 keepsake: a one page analysis report written from your notebook

	Section	What you write	Where it comes from
	**Question**	One sentence naming the target, the predictors and the decision.	The project question slide and your notebook
	**Model**	The multiple regression and the unit of every predictor.	Notebook step 3
	**Findings**	Three coefficients as sentences, with intervals, and R squared.	Notebook step 7: regression_summary.csv and .xlsx
	**Check**	What the residual plot showed and what you would do about it.	Notebook step 4
	**Limits**	What the model cannot prove, and the test RMSE against the baseline.	Notebook step 6

Exit check: three questions, then compare with the short answers

	Question	Short answer
	**The promo_on coefficient is 6.8. Say it as one sentence for a manager.**	A promotion week has about 7 more orders than a week without one at the same footfall and rainfall. It shows an association.
	**A residual plot fans out to the right. What is the name of the problem, and what does it affect?**	Heteroscedasticity. The spread of the errors grows with the prediction, so intervals for large predictions are too narrow.
	**Adding 7 branch indicators moved R squared from 0.795 to 0.796 and adjusted R squared from 0.794 to 0.793. Keep them?**	No. They add almost nothing, and adjusted R squared fell.

Save your notebook, your summary file and your report

What you keep from today

Keep your notebook and regression_summary.csv, and write your one page analysis report: the question, the model, three coefficients as sentences, R squared, the residual check and what the model cannot prove.

Our question today

How much does one thing change when another changes, and how well can a model predict it?

Next session

Session 14: Machine learning, the workflow. We predict loan default with a train and test split.

Take away

Your lab files

Notebook and summary file

Your report

One page, Project milestone 2

Next session

Machine learning